

Invasion in the News: Sentiment, Media Coverage and Signaling in the Russo-Ukraine War

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Outline

- 1 Introduction
 - Research Question
 - Data Overview
 - Results Overview
- 2 Theory
 - State of the Literature
 - The Case
 - Hypotheses
- 3 Data and Methods
- 4 Results
 - Research Question 1
 - Research Question 2
- 5 Takeaways

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- Why do states respond differently to the same event?

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- Differences in Discontinuities Design between Different Country State-Owned Media

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- Finding 3: State media in states with security agreements with Russia respond more negatively than neutral states

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 - Trade as another form of institution, alliance (Keohane and Martin, 1995)
 - Globalization has significantly increased interdependence between states (Keohane and Nye, 1973), (Aloui et al., 2011)

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 - Public Statements are meant to influence other actors (Jervis, 2017)

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 - Public Statements help signal commitments to other states (Fuhrmann and Sechser, 2014)

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 - Non-State actors rely on it as well (Flamer, 2023)

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 - NATO: Albania, Hungary, Turkey
 - CSTO: Azerbaijan, Armenia, Belarus, Kazakhstan
 - Independent: Georgia, Kosovo, Moldova, Serbia
- Invasion of Ukraine in February 2022, ostensibly to support pro-Russia separatists fighting for independent states (Walker, 2024)

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- When the topic is international in nature, states prioritize the international signalling nature of their state-owned media
- Both imposing sanctions and having sanctions imposed are costly (Martin, 1993)
- The stag hunt is a situation where cheap talk is likely to have a positive payoff (Farrell and Rabin, 1996)

Hypotheses - Research Question 1

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- Hypothesis 1A: News sources' coverage of Russia will become more negative even for non-military articles
- Hypothesis 2: State owned media sources' coverage of Russia will respond less negatively than independent media's coverage

Hypotheses - Research Question 2

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- Hypothesis 3 Alternative: State media sentiment is driven by domestic factors.

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- Utilize a differences in discontinuities design to identify the difference in response in different countries' state-owned media

Data Collection

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- Extensive pre-processing, with translation done via transformer (Moratz et al., 2025)

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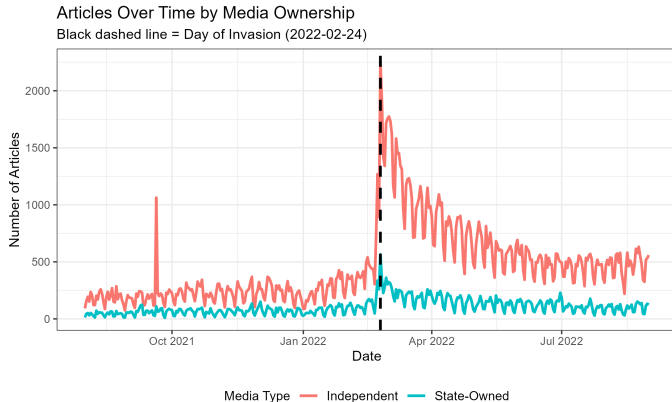
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Summary Statistics

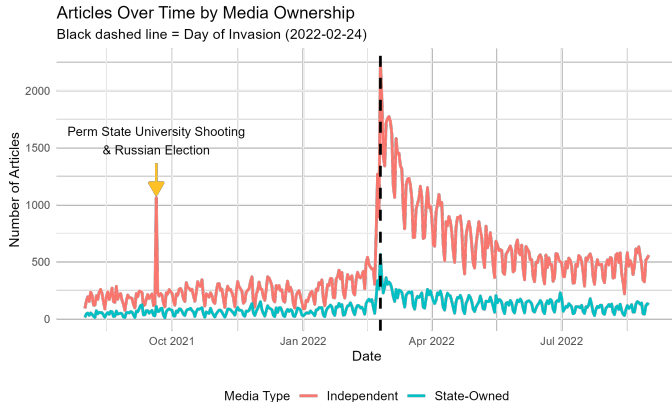
Descriptive Overview of News Article Dataset

Statistic	Value
Total articles	174094
Number of countries	11
Date range	2021-08-01 to 2022-08-31
Mean sentiment	-0.251
Sentiment SD	0.559
State-owned %	22.3
Independent %	77.7
Number of unique sources	62
Number of topics	5

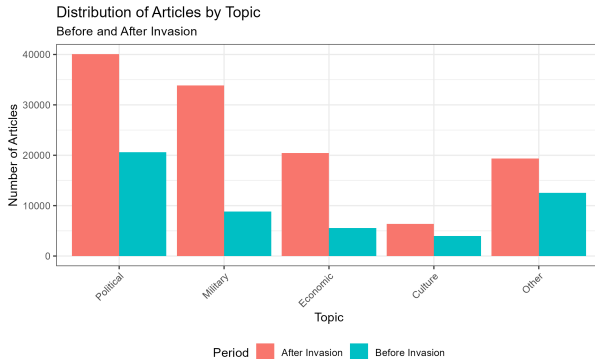
Articles by Day



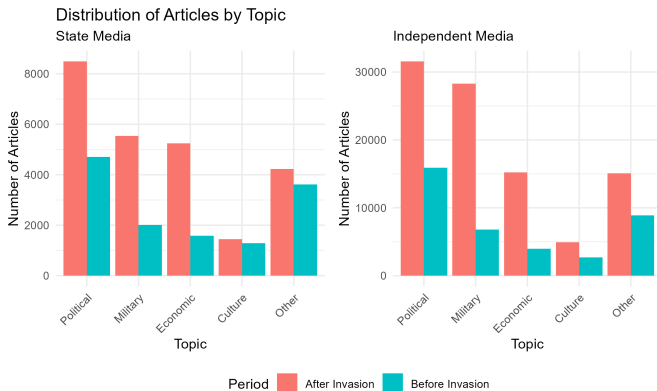
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Topic Chart



Topic Chart by Outlet



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 - Discontinuity of treatment probability
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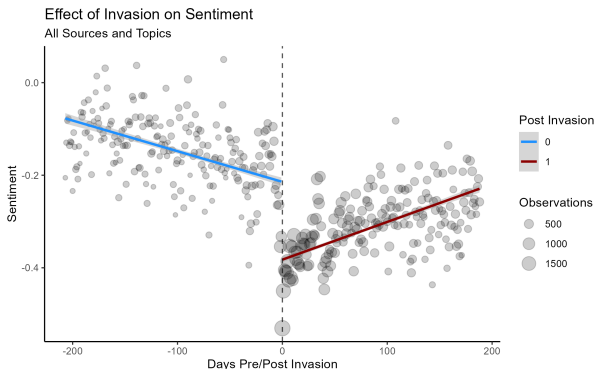
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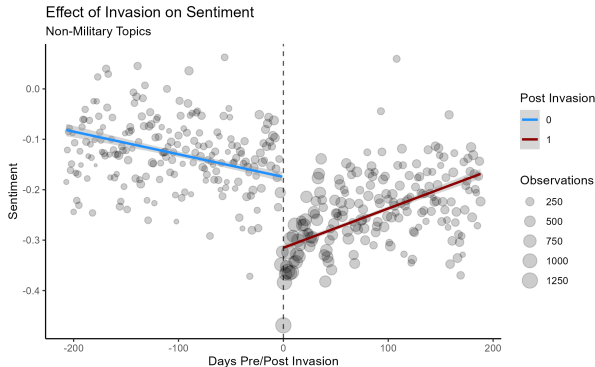
- Research Question 1
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5 Takeaways

Hypothesis 1 - A Decrease in Sentiment



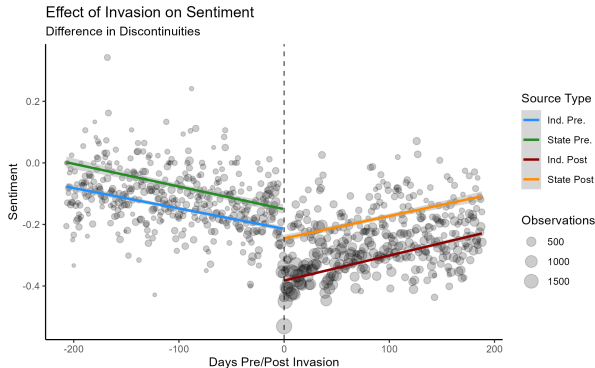
Hypothesis 1A - A Decrease in Sentiment in Non-Military Topics



Hypothesis 1 Table

Difference in Difference Analysis						
	(1)	(2)	(3)	(4)	(5)	(6)
Invasion	-0.168*** (0.012)	-0.158*** (0.011)	-0.157*** (0.011)	-0.1500*** (0.0099)	-0.173*** (0.014)	-0.157*** (0.014)
State FEs	-	X	X	X	X	X
Source FEs	-	-	X	X	X	X
Topic FEs	-	-	-	X	X	X
Num.Obs.	135356	135356	135356	135356	43907	94983
R2	0.025	0.040	0.050	0.074	0.065	0.083
R2 Adj.	0.025	0.040	0.049	0.074	0.064	0.082
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001						
All standard errors are clustered at the source level. Model 2 uses the triangular kernel within the optimal bandwidth. Model 3 omits NATO allies. Model 4 omits military-related articles. Model 5 omits military-related articles and only uses articles that were classified as primarily Russian Related.						

Hypothesis 2 - State vs Independent



Hypothesis 2 Table

	Difference in Discontinuity Analysis				
	(1)	(2)	(3)	(4)	(5)
Invasion	-0.15***	-0.191***	-0.158***	-0.14***	-0.176***
	(0.01)	(0.013)	(0.014)	(0.01)	(0.013)
State Owned x Treatment	0.058**	0.081**	0.081**	0.065***	0.095***
	(0.018)	(0.025)	(0.025)	(0.018)	(0.023)
State FEs	X	X	X	X	X
Source FEs	X	X	X	X	X
Topic FEs	X	X	X	X	X
Num.Obs.	174094	62675	121894	130525	102765
R ²	0.087	0.081	0.099	0.063	0.087
R ² Adj.	0.087	0.080	0.098	0.063	0.086

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Country Level Differences

Mean Sentiment Before and After Invasion			
With Standard Errors and Significance Stars (± 2 Week Window)			
Country	Before	After	Diff (After - Before)
Albania	-0.245 (0.033)	-0.34 (0.028)	-0.095 *
Armenia	-0.055 (0.032)	-0.222 (0.028)	-0.168 ***
Azerbaijan	0.006 (0.024)	-0.035 (0.021)	-0.041
Belarus	-0.035 (0.024)	-0.094 (0.021)	-0.058
Georgia	0.03 (0.134)	0.097 (0.062)	0.066
Hungary	-0.323 (0.038)	-0.325 (0.026)	-0.002
Kazakhstan	0.145 (0.062)	-0.221 (0.053)	-0.366 ***
Kosovo	-0.355 (0.038)	-0.296 (0.04)	0.059
Moldova	-0.301 (0.069)	-0.495 (0.053)	-0.193 *
Serbia	-0.176 (0.078)	-0.181 (0.044)	-0.004
Turkey	-0.307 (0.025)	-0.409 (0.017)	-0.102 ***
Ukraine	-0.41 (0.037)	-0.406 (0.099)	0.004

Hypothesis 3 - Comparison Table

Ethnic Russian Population × Russia Security Alignment

Potential Comparisons

	Low Alignment	High Alignment
Low Ethnic	Kosovo, Serbia	Azerbaijan, Armenia
High Ethnic	Moldova, Georgia	Belarus, Kazakhstan

Hypothesis 3 - International Factors Comparison

Ethnic Russian Population × Russia Security Alignment

Alignment Comparison

	Low Alignment	High Alignment
Low Ethnic	Kosovo, Serbia	Azerbaijan, Armenia
High Ethnic	Moldova, Georgia	Belarus, Kazakhstan

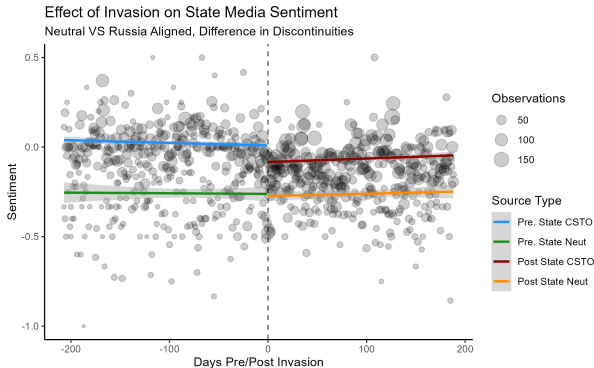
Hypothesis 3 - Domestic Factors Comparison

Ethnic Russian Population × Russia Security Alignment

Ethnic Comparison

	Low Alignment	High Alignment
Low Ethnic	Kosovo, Serbia	Azerbaijan, Armenia
High Ethnic	Moldova, Georgia	Belarus, Kazakhstan

Hypothesis 3 - Alignment Results

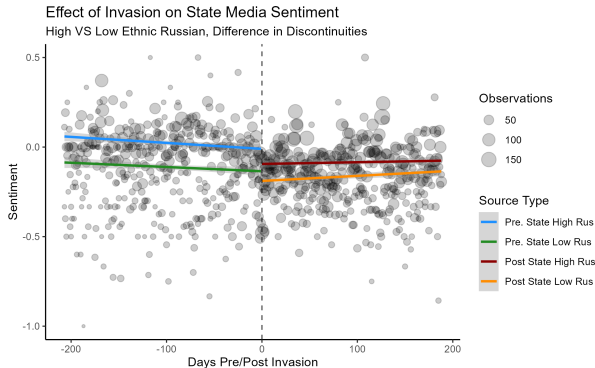


Hypothesis 3 - Alignment Results

Russian Alignment: Difference in Discontinuity Analysis

	(1)	(2)	(3)
Invasion	-0.012	0.015	0.029
	(0.024)	(0.040)	(0.027)
CSTO x Treatment	-0.081**	-0.102*	-0.107***
	(0.028)	(0.048)	(0.031)
Num.Obs.	26911	8878	22535
R2	0.038	0.036	0.033
R2 Adj.	0.038	0.036	0.032
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			
All standard errors are clustered at the source level. Model 2 uses the triangular kernel within the optimal bandwidth. Model 3 omits military-related articles.			

Hypothesis 3 - Alternative



- 1 Introduction
 - Research Question
 - Data Overview
 - Results Overview
- 2 Theory
 - State of the Literature
 - The Case
 - Hypotheses
- 3 Data and Methods
- 4 Results
 - Research Question 1
 - Research Question 2
- 5 Takeaways

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- News outlets respond negatively to the invasion
- This negative sentiment is expressed in a variety of topics, not just those related to military matters
- There is a significant difference in how state-owned media responds compared to independent media - it is significantly less negative
- States with security alignments with Russia respond more negatively to the invasion than neutral states

Thank You!

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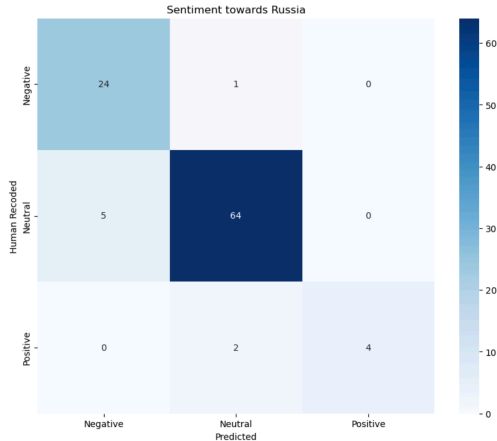
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GPT Sentiment Confusion Matrix



GPT Topic Confusion Matrix

